qe2boltzTwo.py
Version 1.0.2

The following done with Ubuntu 20.04.1 LTS:

```
username@computername: ~/Desktop
username@computername:~/Desktop$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 20.04.1 LTS
Release:        20.04
Codename:       focal
username@computername:~/Desktop$
```

Installed BoltzTraP2 (Version 20.7.1)

1. Installed BolzTraP2 following Google Group posts and/or webpages:
   https://gitlab.com/sousaw/BoltzTraP2
   https://www.imc.tuwien.ac.at/index.php?id=21094

Used Previously Ran Quantum Espresso (QE) Calculation

2. Obtained BoltzTraP_Tutorial.nscf.out from a previously ran QE calculation:

```
username@computername: ~
username@computername:~$ wget http://levilentz.com/BoltzTraP/BoltzTrap_Tutorial.zip
```

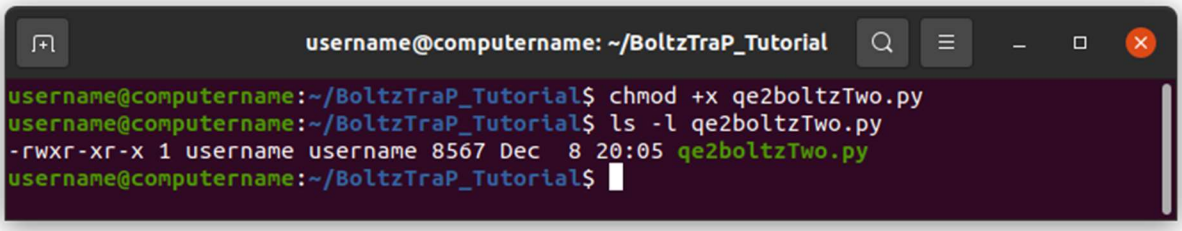
```
gavin@gavincomputer: ~
gavin@gavincomputer:~$ unzip BoltzTrap_Tutorial.zip
```

```
username@computername: ~/BoltzTraP_Tutorial
username@computername:~$ cd BoltzTraP_Tutorial/
username@computername:~/BoltzTraP_Tutorial$ ls -l *.nscf.out
-rw-rw-r-- 1 username username 83431 Feb 21  2015 BoltzTraP_Tutorial.nscf.out
username@computername:~/BoltzTraP_Tutorial$
```

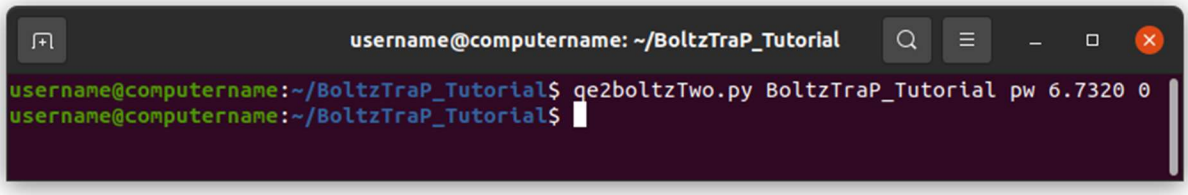
Running QE calculation references:
https://blog.levilentz.com/boltztrap-tutorial-for-quantum-espresso/
http://www.fisica.uniud.it/~giannozz/QE-Tutorial/handson_pwscf.pdf
http://levilentz.com/BoltzTraP/BoltzTrap_Tutorial.zip

Installed and Ran qe2boltzTwo.py Script

3. Place qe2boltzTwo.py in the current calculation directory (or it could be placed in the QE installation bin folder [e.g., ~/q-e-qe-6.6/bin]):



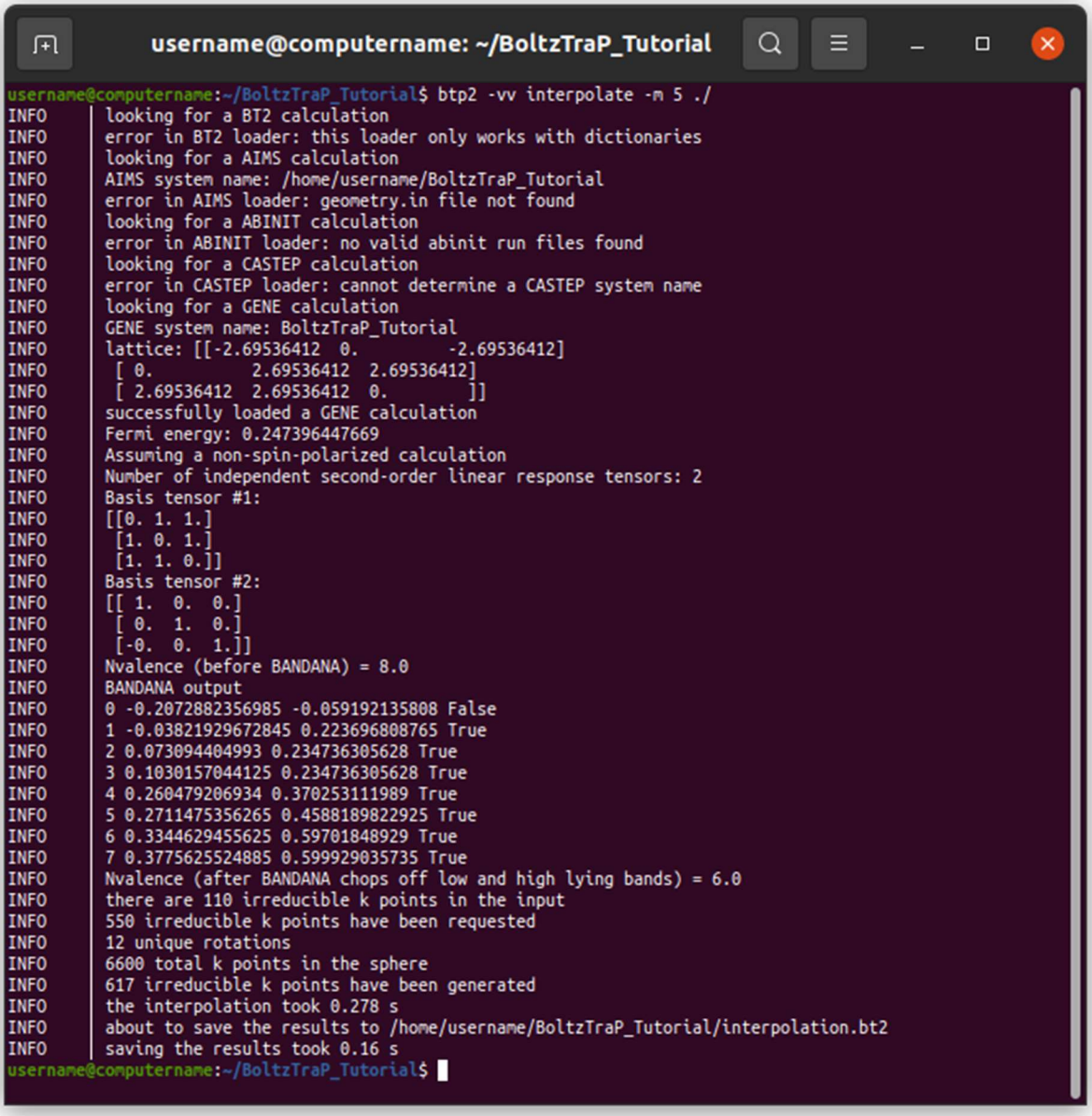
4. Ran QE to BoltzTraP2 script for file conversion:



Ran BoltzTraP2 Calculation

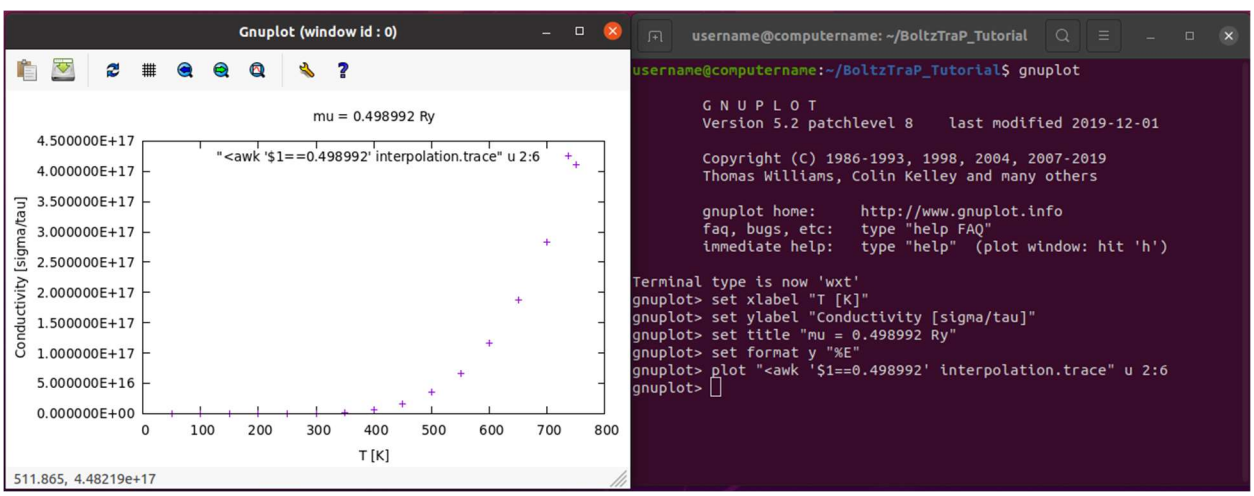
References:
https://gitlab.com/sousaw/BoltzTraP2/wikis/tutorial
https://www.researchgate.net/post/how_to_grep_trace_file_of_BoltzTrap

1. Ran BoltzTraP2 interpolate:



```
username@computername: ~/BoltzTraP_Tutorial
username@computername:~/BoltzTraP_Tutorial$ btp2 -vv interpolate -m 5 ./
INFO     | looking for a BT2 calculation
INFO     | error in BT2 loader: this loader only works with dictionaries
INFO     | looking for a AIMS calculation
INFO     | AIMS system name: /home/username/BoltzTraP_Tutorial
INFO     | error in AIMS loader: geometry.in file not found
INFO     | looking for a ABINIT calculation
INFO     | error in ABINIT loader: no valid abinit run files found
INFO     | looking for a CASTEP calculation
INFO     | error in CASTEP loader: cannot determine a CASTEP system name
INFO     | looking for a GENE calculation
INFO     | GENE system name: BoltzTraP_Tutorial
INFO     | lattice: [[-2.69536412  0.         -2.69536412]
INFO     |  [ 0.          2.69536412  2.69536412]
INFO     |  [ 2.69536412  2.69536412  0.        ]]
INFO     | successfully loaded a GENE calculation
INFO     | Fermi energy: 0.247396447669
INFO     | Assuming a non-spin-polarized calculation
INFO     | Number of independent second-order linear response tensors: 2
INFO     | Basis tensor #1:
INFO     | [[0. 1. 1.]
INFO     |  [1. 0. 1.]
INFO     |  [1. 1. 0.]]
INFO     | Basis tensor #2:
INFO     | [[ 1.  0.  0.]
INFO     |  [ 0.  1.  0.]
INFO     |  [-0.  0.  1.]]
INFO     | Nvalence (before BANDANA) = 8.0
INFO     | BANDANA output
INFO     | 0 -0.2072882356985 -0.059192135808 False
INFO     | 1 -0.03821929672845 0.223696808765 True
INFO     | 2 0.073094404993 0.234736305628 True
INFO     | 3 0.1030157044125 0.234736305628 True
INFO     | 4 0.260479206934 0.370253111989 True
INFO     | 5 0.2711475356265 0.4588189822925 True
INFO     | 6 0.3344629455625 0.59701848929 True
INFO     | 7 0.3775625524885 0.599929035735 True
INFO     | Nvalence (after BANDANA chops off low and high lying bands) = 6.0
INFO     | there are 110 irreducible k points in the input
INFO     | 550 irreducible k points have been requested
INFO     | 12 unique rotations
INFO     | 6600 total k points in the sphere
INFO     | 617 irreducible k points have been generated
INFO     | the interpolation took 0.278 s
INFO     | about to save the results to /home/username/BoltzTraP_Tutorial/interpolation.bt2
INFO     | saving the results took 0.16 s
username@computername:~/BoltzTraP_Tutorial$
```

2. Ran BoltzTraP2 integrate:

```
username@computername: ~/BoltzTraP_Tutorial
username@computername:~/BoltzTraP_Tutorial$ btp2 -vv integrate interpolation.bt2 -b 5000 50:800:50
INFO     | looking for a BT2 calculation
INFO     | successfully loaded a BT2 calculation
INFO     | Fermi energy: 0.247396447669
INFO     | sucessfully loaded interpolation.bt2
INFO     | rebuilding the bands took 1.22 s
INFO     | computing the DOS took 0.39 s
INFO     | smoothing the DOS took 0.152 s
INFO     | Temperatures: [ 50. 100. 150. 200. 250. 300. 350. 400. 450. 500. 550. 600. 650. 700.
INFO     |  750.]
INFO     | Fermi level from DFT: 0.247396447669
INFO     | Refined Fermi level: 0.249495958456657
INFO     | Intrinsic chemical potential for each T: [0.24949596 0.24949596 0.24949596 0.24949596 0.24949596 0
.24949596
INFO     |  0.24949596 0.24949596 0.24949596 0.24949596 0.24949596 0.24725303
INFO     |  0.24717632 0.24710877 0.24704592]
INFO     | Chemical potentials: [-0.02198354 -0.02185315 -0.02172275 ...  0.58682423  0.58695462
INFO     |   0.58708502]
INFO     | computing the FD moments took 238 s
INFO     | trace output file: interpolation.trace
INFO     | conductivity/seebeck output file: interpolation.condtens
INFO     | Hall coefficient output file: interpolation.halltens
INFO     | JSON output file: interpolation.btj
username@computername:~/BoltzTraP_Tutorial$
```

3. Plot of temperate (x-axis) versus sigma/tau (y-axis):